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Hericium erinaceus polysaccharide-protein HEG-5 inhibits SGC-7901 cell growth *via* cell cycle arrest and apoptosis

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Abstracts

HEG-5 is a novel polysaccharide-protein purified from the fermented mycelia of *Hericium erinaceus* CZ-2. The present study aims to investigate the effects of HEG-5 on proliferation, cell cycle and apoptosis of human gastric cancer cells SGC-7901. Here, we first uncover that HEG-5 significantly inhibited the proliferation and colony formation of SGC-7901 cells by promoting apoptosis and cell cycle arrest at S phase. RT-PCR and Western blot analysis suggested that HEG-5 could decrease the expressions of Bcl2, PI3K and AKT1, while increase the expressions of Caspase-8, Caspase-3, p53, CDK4, Bax and Bad. These

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findings indicated that the Caspase-8/-3-dependent, p53-dependent mitochondrial-mediated and PI3k/Akt signaling pathways involved in the molecular events of HEG-5 induced apoptosis and cell cycle arrest. Thus, our study provides *in vitro* evidence that HEG-5 may be taken as a potential candidate for treating gastric cancer.

KEYWORDS: *Hericium erinaceus; Polysaccharide-protein; Anti-tumor activity; Cell cycle; Apoptosis*

1. Introduction

Mushrooms are ubiquitous in nature and have high nutritional/pharmaceutical attributes. Many species such as *Ganoderma lucidum* and *Hericium erinaceum* have long been used as traditional Chinese medicines or functional foods in China, Japan and other Asian countries for the treatment of gastroenteric disorder, lymphatic disease and various cancers [1]. In particular, low-molecular-weight (e.g. cerebrosides, isoflavones, catechols, amines, triacylglycerols, sesquiterpenes and steroids) and high-molecular-weight (e.g. polysaccharides, polysaccharide-proteins/peptides and proteins) compounds are considered as the major functional constituents responsible for mushrooms' biological activities [2, 3, 4, 5].

Generally, polysaccharide-proteins/peptides complexes have different structures and bioactivities from those of polysaccharides in mushrooms. Polysaccharides possess the pure carbohydrate chains and are considered as biological response modifiers (BRMs) and used as the nonspecific immune-stimulator for cancer treatment. For example, the homogeneous *Grifola frondosa* polysaccharide GFPBW2 and *Phellinus igniarius* extracellular

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polysaccharides (EPS) showed their antitumor activities by activating the macrophages or balancing the numbers of White blood cells (WBC), Red blood cell (RBC), platelet (PLT) and Hemoglobin (HGB) *in vitro* and *in vivo* [6,7]. However, polysaccharide-proteins/peptides complexes contain the protein/polypeptide covalently and specifically bound with carbohydrates as the side-chains [8, 9]. In most cases, polysaccharide-proteins/peptides inhibit the tumor cells growth *in vitro* by activating multiple signal pathways including cell cycle arrest, DNA damage, and alteration of death inhibitors/promoters expression [10]. Previous studies had revealed that *Ganoderma lucidum* glycopeptide GLPS-SF1, *Grifola frondosa* polysaccharide-peptide GFPS1b, *Pleurotus citrinpileatus* glycoprotein PCP-3A and pea glycoprotein could kill the tumor cells directly *via* inducing cell cycle arrest, up-regulating *Bax* gene expression and/or down-regulating *Bcl-2*/*Bcl-xl* genes expression [11, 12, 13, 14].

The edible mushroom *Hericium erinaceus* belongs to the order *Aphyllophorales* and family *Hycnaecae* [15]. Some biological components such as polysaccharides, lectins, amycenone, and aromatic compounds have been obtained from *H. erinaceus* mycelia or fruiting body and showed the pharmacological properties [16, 17, 18, 19]. In our previous study, a hemagglutinating/antitumor polysaccharide-protein (glycoprotein) HEG-5 with the novel and thermo-unstable structure was purified from the *H. erinaceus* fermented mycelia [20, 21, 22]. To further elucidate the mechanism of HEG-5 on inhibiting tumor cells, the present study aims to evaluate the inhibitory effect of HEG-5 and its role on cell cycle arrest and apoptosis of SGC-7901 cells, and reveal a possible signal pathway for elucidating the HEG-5' antitumor molecular mechanisms.

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2. Materials and methods

2.1. Materials

All the cell culture reagents were purchased from Hyclone laboratories (South Logan, Utah, USA). Hoechst dye 33258, agrose, ethidium bromide (EtBr), 3-(4,5-Dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT), dimethyl sulfoxide (DMSO), propidium iodide (PI) and Annexin V/FITC assay kit were purchased from Sigma Chemicals (St. Louis, MO). RNAiso Plus, PrimeScript^{TM} RT reagent Kit with gDNA Eraser, and SYBR *Premix Ex*Taq^{TM} were obtained from Takara Bio (Kyoto, Japan). Polymerase chain reaction (PCR) primers were custom-synthesized and purified by Sangon Biotech Co., Ltd (Shanghai, China). The primary antibodies for Caspase-3, Caspase-8, Bcl2, Bad and p53, and Goat Anti-Rabbit IgG H&L second antibody (HRP) were purchased from Abcam Company (Cambridge, UK). All other chemicals were analytical grade.

2.2. Isolation and purification of polysaccharide-protein HEG-5

Hericium erinaceus polysaccharide-protein HEG-5 was extracted and purified based on a method described previously [22]. Briefly, *H. erinaceus* fermented mycelia were blended with 1L distilled water for 1 min and then extracted at 4 ^{o}C for 24 h. After centrifugation at 5,030 × *g* for 20 min, the supernatant was precipitated with 80% saturated ammonium sulphate overnight at 4 ^{o}C, and centrifuged (12,864 × *g*, 20 min). The precipitate was collected, then re-dissolved and dialyzed against distilled water for 3 days. HEG-5 was obtained by separating the water-soluble crude proteins on a DEAE-Sepharose fast-flow

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anionic resin column (Pharmacia AP, GE) eluted sequentially with 0.1, 0.3 and 0.5 M NaCl solution at pH 7.0 and Superdex^{TM} 75 prep grad column (Amersham Biosciences, Sweden) eluted with distilled water using an AKTA purifier 100 (Amersham Biosciences, Sweden).

2.3. Cell Cultures

Human gastric carcinoma cell line (SGC-7901) was purchased from Type Culture Collection of Chinese Academy of Sciences (Shanghai, China). The cells were maintained in RPMI-1640 medium supplemented with 10 % fetal bovine serum (FBS), and 1 % P/S (100 units penicillin and streptomycin /mL) at 37 ^{o}C in a 5 % carbon dioxide atmosphere.

2.4. MTT Assay

The viability of cells was determined with MTT assay. SGC-7901 cells in their exponential growth phase were plated into a 96-well culture plate at the density of 5×10^4 cells per well and incubated for 24 h. Then, the cells were exposed to 25, 50, 100 and 200 μg/mL of HEG-5 for 24 h and 48 h. The absorbance was measured at 570 nm in the microplate reader (Thermo, USA). The inhibition ratio was calculated as follows:

Inhibition ratio (%) = [OD_{570nm} (control)-$OD_{570\ nm}$ (sample)]/$OD_{570\ nm}$ (control) × 100.

2.5. Morphological studies of apoptotic cell

2.5.1. Light Microscopy

SGC-7901 cells were plated in 6-well plates at the density of 5×10^4 cells per well for 12 h, then treated with different concentrations of HEG-5. After 48 h, the cells were observed

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under inverted light microscope.

2.5.2. Scanning Electron Microscopy

HEG-5-treated SGC-7901 cells were washed three times with PBS and pre-fixed with 2.5 % (v/v) glutaraldehyde at 4 ^{o}C for 1 h. The pre-fixed cells were rinsed thoroughly in PBS, post-fixed in 1% OsO_4 at 4 ^{o}C for 1 h, and dehydrated at increasing concentrations of ethanol (60 %, 70 %, 80 %, 90 % and 100 %, v/v). The surface morphology of HEG-5-treated SGC-7901 cells was observed under scanning electron microscope (SEM) at magnification of 400 times.

2.5.3. Hoechst Staining Observation

The cell nucleus shape was observed by fluorescence microscope following Hoechst 33258 staining [23]. SGC-7902 cells were exposed to different concentrations of HEG-5 for 48 h, then washed twice with PBS, fixed by methanol pre-cooled at -20 ^{o}C, and stained using Hoechst dye 33258 for 5 min at room temperature. A DMI 4000B fluorescent microscope (LEICA, Germany) was applied to observe the nuclear shape of HEG-5-treated SGC-7901 cells.

2.6. Comet Assay

SGC-7901 cells in their exponential growth phase were plated into a 96-well culture plate at the density of 1×10^5 cells per well and were exposed to HEG-5 (0, 20, 50, and 80 μg/mL) for 48 h. The cells were collected, rinsed and suspended in PBS. Cells suspension (30 μL, 1×10^4 cells) was mixed with 110 μL of 1 % (v/v) molten low-melting-point agarose at 37

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^{o}C. The mono-suspension was cast on a microscopic slide covered with a layer of 0.8 % regular-melting-point agarose. The agarose was gelled at 4 ^{o}C for 20 min and the slide was then immersed in a cold, freshly prepared lysing solution (2.5 M NaCl, 100 mM EDTA, 10 mM Tris-HCl, 1 % Triton-X 100, 10 % DMSO, pH 10) for 30 min at 4 ^{o}C. After lysis, the slides were washed three times using distilled water and immersed in fresh alkaline electrophoresis solution contained 300 mM NaOH, 1mM EDTA, pH 13 for 10 min at 4 ^{o}C. Electrophoresis was carried out in ice bath (4 ^{o}C) at 300 mA and 25 V for 20 min. The slides were neutralized to pH 7.5 in a 0.4 M Tris buffer, and stained with 50 μL of 20 μg/mL ethidium bromide solution. The stained slides were then analyzed using a DMI4000B fluorescent microscope (LEICA, Germany) with a 515-560 nm excitation filter and a 590 nm barrier filter. Tail moment was used to estimate the DNA damage (arbitrary units).

2.7. Analysis of Apoptotic Cells using Annexin V-FITC and PI Double Staining

SGC-7901 cells (5×10^5 cells/mL) were seeded into 6-well plates. The cells were treated with HEG-5 for 48 h, collected and washed twice with cold PBS, and then re-suspended in the 500 μL of 1×binding buffer with addition of 5 μL of Annexin V-FITC Conjugate and 10 μL of Propidium Iodide (PI) Solution for 10 min. After staining, the fluorescence of the cells was determined immediately using a flow cytometry (Becton Dickinson, Mountain View, CA). Cells collected at the early stage of apoptosis were stained using the Annexin V-FITC Conjugate alone. Live cells will not be stained by either the PI or Annexin V-FITC Conjugate. Necrotic cells will be stained by both the PI and Annexin V FITC Conjugate.

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2.8. Cell Cycle Distribution Assay

SGC-7901 cells (5×10^5 cells/mL) were seeded into 6-well plates and cultured for 12 h. The cells were exposed to different concentrations of HEG-5 for 48 h, collected, washed twice, and re-suspended with cold PBS. These cells were fixed with 500 μL of 70 % alcohol for 24 h at 4 ^{o}C, and then washed twice with cold PBS. Finally, the cells were stained with a solution consisted of 100 μg/mL PI and 100 μg/mL RNase A and incubated for 30 min at 4 ^{o}C. A FACS Flow cytometer was applied to analyze the cell cycles.

2.9. Gene Expression analysis Using qRT-PCR

Cells (1.5×10^6 cells/mL) were plated in 25 cm^2 T-flasks for 12 h, and then treated with different concentrations of HEG-5 for 48 h. The cells were collected and total RNA was extracted by an RNAiso Plus kit. Reverse transcription was carried out using PrimeScript^{TM} RT reagent Kit with gDNA Eraser. The cDNA samples were incubated with SYBR *Premix Ex*Taq^{TM} kit and the expressed genomes were analyzed using the BioRad. The primer sequences used for amplification of target human genes are listed in Table 1. The target mRNA expression were calculated by using the ΔΔCt method. Fold change values were presented as $2^{-\Delta\Delta Ct}$ for the genes relative to the control sample [24]. All the PCR reactions were analyzed in DNA-PAGE using 2×TBE as the buffer. DNA ladder 2000 was used as the marker.

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2.10. Western Blot analysis

The cells were washed with ice-cold PBS and lysed directly using ice for 30 min in 400 μL of cold lysis buffer (150 mM NaCl, 50 mM of pH 7.4 Tris, 1 mM EDTA, 1 % Triton X-100, 0.5 % SDS, 0.01 % PMSF). The cell lysates were centrifuged at 12,000 $\times g$ for 15 min and the supernatants were collected. The protein concentrations were determined using a BCA protein assay kit (BestBio, Shanghai, China). Thirty micrograms of protein samples were separated in a 12 % SDS-polyacrylamide gel and transferred to a polyvinylidenedifluoride (PVDF) membrane in the TBST buffer (1 M Tris-HCl, 1.5 M NaCl, 20 % Tween20). The membranes were blocked in 5 % nonfat dry milk in TBST for 2 h, washed three times by TBST, and probed with primary antibodies at 4 ^{o}C for 12 h. Membranes were then incubated with horseradish peroxidase-conjugated goat anti-rabbit secondary antibody [1:4000] for 2.5 h. Following a wash with TBST, membranes were developed with ECL detection reagent; bands were visualized on X-ray film (Version 1.43, National Institutes of Health, USA). The target protein amounts were normalized to β-actin levels using densitometry. The relative fold changes in protein levels were then calculated.

2.11. Statistical Analysis

All results were presented as mean ± standard deviation in triplicate (n=3). Differences between groups were considered to be significant at $p < 0.05$.

3. Results

3.1 HEG-5 exerts growth inhibition effects on SGC-7901 cells

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MTT results show that HEG-5 significantly inhibited the SGC-7901 cells proliferation (P<0.05) (Fig. 1). The highest inhibitory rate of HEG-5 was 93.4 % when its concentration increased to 200 μg/mL after 48-h treatment. The IC_{50} values at 24 h and 48 h were calculated to be 107.4 μg/mL and 46.7 μg/mL, respectively.

3.2 HEG-5 induces the morphological changes of SGC-7901 cells

SGC-7901 cells treated with HEG-5 (0, 20, 50, and 80 μg/mL) for 48 h were directly observed under an inverted light microscope (Fig. 2). With increasing concentrations of HEG-5, the number of cells having unaltered morphology gradually decreased. Additionally, the cell morphology began to change as round-shaped cells shrinkage and detachment from culture plate. SEM figures (Fig. 3) showed that the untreated cells having unaltered morphology had numerous intact folds, while HEG-5-treated SGC-7901 cells showed the typical apoptotic characteristics including loss of folds, appearance of blebbing, and formation of apoptotic body.

3.3 HEG-5 promotes apoptosis of SGC-7901 cells

Nuclear morphological changes of SGC-7901 cells treated with or without HEG-5 were confirmed by Hoechst 33258 staining (Fig. 4). Compared to the round and homogeneously stained nuclei in the control group, the nucleus of HEG-5 treated SGC-7901 cells were fragmented, which demonstrated the typical characteristics of apoptosis. The alkaline comet assay presented the shapes of DNA fragments migration (Fig. 5). The untreated SGC-7901 cells had the comet heads with high-density DNA accompanied with the smooth margin and

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intact nuclei. After treatment of HEG-5 at 50 μg/mL and 80 μg/mL, the clear comet-like tails of DNA migration could be observed.

The HEG-5-induced apoptosis levels were confirmed and quantified by FACS analysis after staining with Annexin V/PI. From Fig. 6, the untreated cells showed normal cell viability without significant cell apoptosis. When SGC-7901 cells were treated with 50 and 80 μg/mL of HEG-5, the percentages of early apoptotic cells significantly raised up to 11.2 % and 16.4 % in contrast with control cells ($P<0.05$), while those of late/necrotic apoptotic cells increased to 24.6 % and 35.8 %. The results suggested that HEG-5 induced cell apoptosis followed in a dose-dependent manner.

3.4 HEG-5 facilitates cell cycle arrest of SGC-7901 cells

To determine whether the growth inhibitory effect of HEG-5 was due to cell cycle arrest, we investigated cell cycle phase distribution of SGC-7901 cells treated with HEG-5 for 48 h by flow cytometry. The relative percentages of cells staying at the G0/G1, S and G2/M phases in control group were 65.48 %, 27.67 % and 6.85 %, respectively. After treatment with 20, 50 and 80 μg/mL of HEG-5, the S phase population increased to 63.97 %, while the relative percentages of cells staying at G0/G1 and G2/M phase decreased to 35.0 % and 1.03 %, respectively (Fig. 7). These results demonstrate that HEG-5 caused cell cycle arrest at the S phase.

3.5 Effects of HEG-5 on genes expressions related to cell cycle and apoptosis

Expression differences in genes involved in the cell cycle and apoptosis were analyzed

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by qRT-PCR after HEG-5 treatment. As shown in Fig. 8 and 9, HEG-5 could significantly increase the mRNA expression of p53, Bad, Bax, Caspase 8, and Caspase 3, and decreased the mRNA expression of CDK4, Bcl 2, PI3K, and Akt in a dose- and time-dependent manner ($p<0.05$). After treated with HEG-5 at the concentration of 50 μg/mL, the CDK4, Bcl 2, PI3K and Akt gene expressions were down-regulated by 0.608, 0.835, 0.486 and 0.889-fold, while the p53, Bad, Bax, Caspase 8, and Caspase 3 mRNA levels increased by 2.085, 2.969, 1.591, 2.462 and 2.549 folds.

3.6 Effects of HEG-5 on proteins expressions related to apoptosis

Fig. 10 showed that HEG-5 treatment significantly up-regulated the levels of p53, Bax, Caspase 8 and Caspase 3 in SGC-7901 cells ($p<0.05$), whereas the Bad protein expression was not significant ($p>0.05$). After treatment with HEG-5 at 50 μg /mL, the p53, Bax, Caspase 8 and Caspase 3 proteins levels were increased by 1.622, 2.034, 1.951, and 1.341 folds, respectively.

4. Discussion

International Agency for Research on Cancer in 2012 reported that gastric cancer is the third leading cause of cancer-related deaths (0.7 millions, 8.8 % of the total) in worldwide [25, 26]. The American Cancer Society also provided a stomach cancer statistics that about 22,220 cases of stomach cancer were diagnosed (13,730 in men and 8,490 in women), and approximately 10,990 people would die from this type of cancer (6,720 men and 4,270 women) in the United States. Some traditional chemotherapeutic agents for gastric cancer

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treatment showed unbearable toxicity or side effects [27, 28]. Thus, to exploit the natural and less toxic antitumor agents appeared to be of remarkable importance. Recently, many edible/medical mushrooms have been proven as the resources for screening these antitumor compounds. Our previous study had firstly clarified the structural characterization and predicted the 3D model of the novel polysaccharide-protein HEG-5 purified from *H. erinaceus* fermented mycelia [22]. Here, the obtained data (Fig.1) revealed that HEG-5 could significantly inhibit the proliferation of SGC-7901 cells in a dose and time-dependent manner with IC_{50} of 107.4 μg/mL at 24 h and 46.7 μg/mL at 48 h, which are in accordance with those of other polysaccharide-proteins/peptides. For example, the polysaccharide-proteins isolated from *Solanum nigrum* Linne and the brown alga *Laminaria japonica* had been found to show the cytotoxic effects on HCT-116, AGS, HepG2 or HT-29 cell lines in a dose-dependent manner [29, 30].

Apoptosis had the typical characteristics of morphological alterations such as membrane bledding, cell shrinkage, chromatin condensation, and nuclear fragmentation with formation of apoptosis bodies [31]. In the present study, several typical morphologies of cell apoptosis such as significant increase of the chromatin condensation and comet tails were observed in the HEG-5 treated SGC-7901 cells (Fig. 3 and 4). The similar phenomena were also found in the Aflatoxin G_1 treated human cancer cell line A549 [32].

Caspases are situated at pivotal junctions in apoptosis pathways. Particularly, Caspase-3 is activated by Caspase-8 and -9 and amplifies initiation signals into full-fledged commitment to disassembly [33]. Fig. 8, 9 and 10 presented quantitative analysis of Caspase-3 and

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Caspase-8 by using qRT-PCR and western blot, which indicates HEG-5 up-regulated Caspase -8 and -3 expressions in SGC-7901 cells in a dose- and time-dependent manner. Similar results could be found that mycelia and culture supernatants of *Cordyceps militaris* induced the apoptosis in human glioblastoma GBM8401 cells through activation of Caspase-3 and Caspase-8 [34].

Mitochondria are regarded as the main organelle for relaying the signal cascade of cell apoptosis. The mitochondrial function is regulated through the Bcl-2 family proteins comprising anti-apoptotic (*Bcl-2*, *Bcl-xl*) and pro-apoptotic genes (*Bax*, *Bad*) [35]. Activation of Bad prevents anti-apoptotic proteins Bcl-2 and Bcl-xl, which further promotes pro-apoptotic protein Bax, and leads to cell death. In this study, HEG-5 treatment resulted in a significant increase of Bad and Bax expression and a decrease of Bcl-2 and Bcl-xl, suggesting that changes in the ratio of pro-apoptotic and anti-apoptotic Bcl-2 family proteins might contribute to the HEG-5 induced apoptosis. Additionally, tumor suppressor p53 is closely related to DNA repair and cell cycle arrest, and induces the apoptosis by directly suppressing Bcl-2 and activating Bax [36]. Herein, enhancement of p53 expression indicated its involvement in HEG-5-induced apoptosis.

The inhibited PI3K/Akt pathway produces mitochondria-mediated apoptosis in various tumor cell lines such as human breast cancer MDA-MB-231 [37]. The effector Akt can inhibit the conformational change of the pro-apoptotic Bax protein and its translocation to mitochondria. Our present study revealed that HEG-5 inhibited the PI3K and Akt expression and induced the following SGC-7901 cell death. These results were in line with the apoptosis

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induced by the compounds such as Chicoric acid through ROS-Mediated PI3K/Akt and MAPK signaling pathways [36].

Cell cycle arrest represents a major regulatory mechanism for cell growth. Most of the antitumor agents can block the progression of cell cycle, which is partly controlled by a family of protein kinase complexes in eukaryotic cell including cyclin-dependent kinase (CDKs) [38]. FACS and Western blot results (Fig. 7) proved that HEG-5 arrested SGC-7901 cells at S phase by decreasing the CDK4 expression level.

In conclusion, we have firstly demonstrated that the polysaccharide-protein HEG-5 from *H. erinaceus* fermented mycelia had the potent inhibited activity to human gastric cancer line SGC-7901 in a dose and time dependent manner. The Caspase-8/-3 dependent, mitochondrial-mediated, and PI3K/Akt pathways are mainly involved in the HEG-5 induced cell cycle arrest and apoptosis (Fig. 11). These findings would contribute to understand the tumor-inhibitory mechanism of polysaccharide-protein HEG-5. Further *in vivo* experiments are ongoing in our lab to evaluate the safety and efficacy of the polysaccharide-proteins/peptides for cancer treatment.

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Table 1. Sequence of primers used in quantitative RT-PCR

Target gene	primer	Nucleotide sequence
p53	F	CACGTACTCTCCTCCCCTCAATA
	R	TCTTCCAGTGTGATGATGGTAAGG
CDK4	F	GAGGCGACTGGAGGCTTTT
	R	GGATGTGGCACAGACGTCC
Bcl 2	F	GGATTGTGGCCTTCTTTGAG
	R	CAGCCAGGAGAAATCAAACAG
Bax	F	TCCACCAAGAAGCTGAGCGA
	R	GTCCAGCCCATGATGGTTCT
Bad	F	CGAGTGAGCAGGAAGACTCC
	R	CTGTGCTGCCCAGAGGTT
Caspase 3	F	AACCTCAGGGAAACATTCAG
	R	GGCTCAGAAGCACACAAAC
Caspase 8	F	GGATGCCTTGATGTTATTCC
	R	AGTTCCCTTTCCATCTCCTC
AKT1	F	GGCGAGCTGTTCTTCCACCT
	R	ATTGTCCTCCAGCACCTCGG
PI3K	F	ATTCCCAGTCAGAGGCGCTAT
	R	GAACTTGTCTTCCCGTCGTGT
β-Actin	F	GTCCACCGCAAATGCTTCTA
	R	TGCTGTCACCTTCACCGTTC

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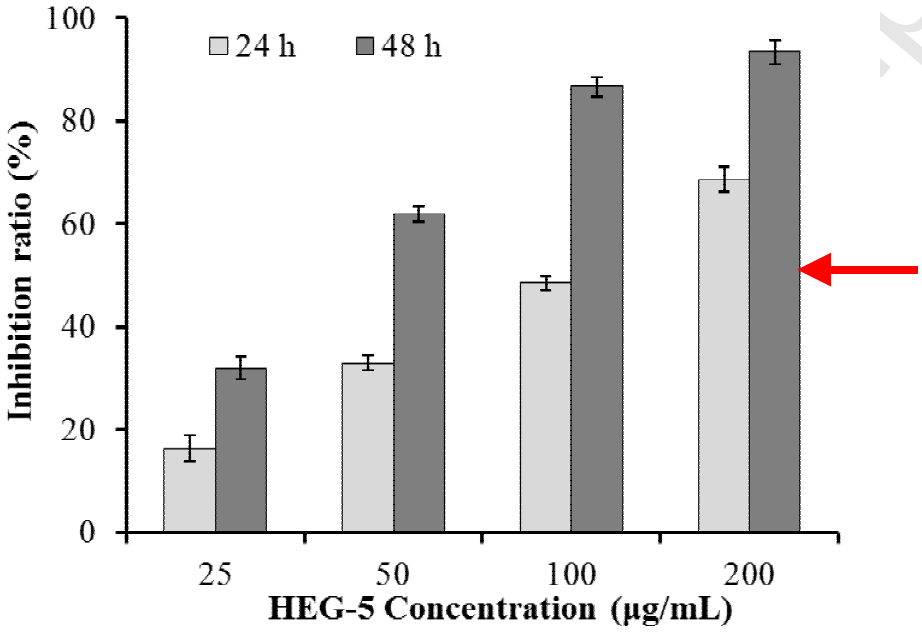

Fig. 1. Growth inhibitory effects of HEG-5 on SGC-7901 cells. SGC-7901 cells in their exponential growth phase were plated into a 96-well culture plate at the density of 5×10^4 cells per well and incubated for 24 h. Then, the cells were exposed to 25, 50, 100 and 200 μg/mL HEG-5 sample for 24 h and 48 h. The effects of HEG-5 on the growth of SGC-7901 cells were determined by MTT assay. Data represent the mean ±SD of three replicates.

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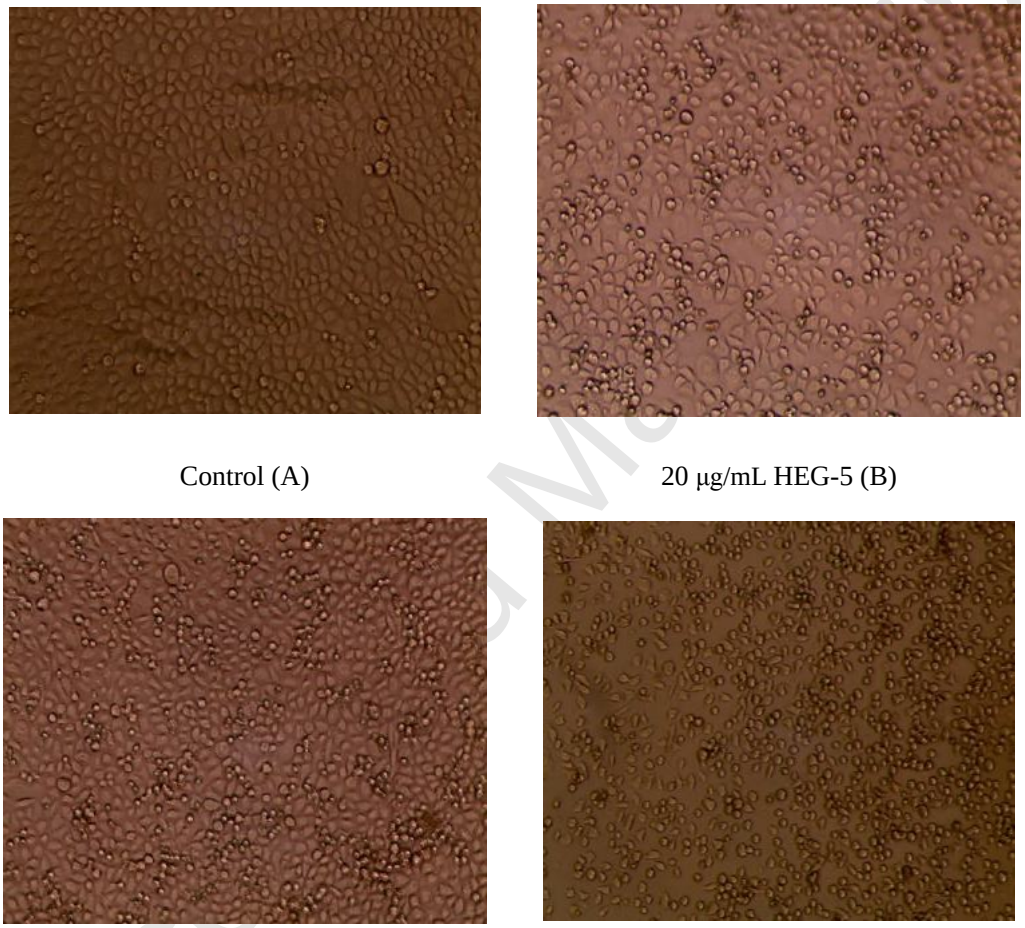

Fig. 2. Inverted light micrographs of SGC-7901 showing morphological changes of cells treated without (A) and with 20 μg/mL (B), 50 μg/mL (C), and 80 μg/mL (D) of HEG-5 for 48 h (×10). With increasing concentrations of HEG-5, the number of normal cells gradually decreased. Additionally, the cell morphology began to change as round-shaped cells shrinkage and detachment from culture plate.

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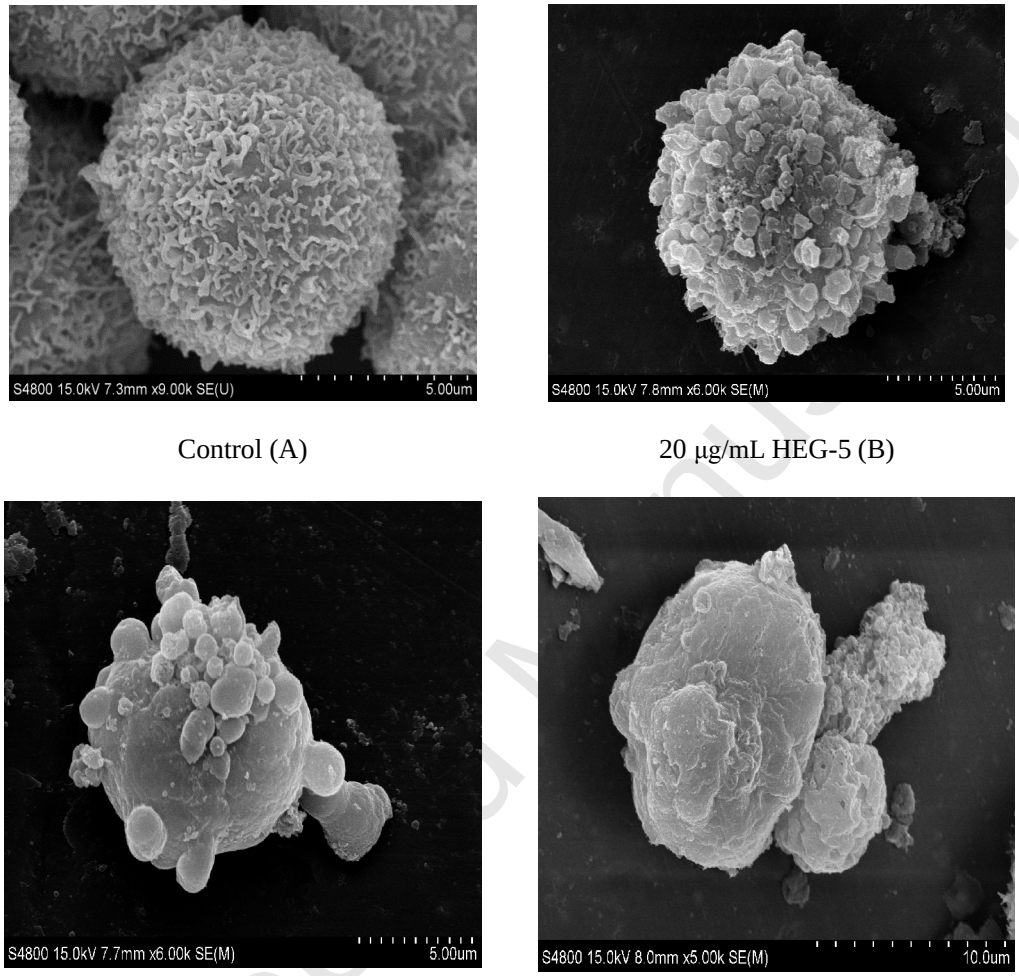

Control (A) 20 μg/mL HEG-5 (B)

50 μg/mL HEG-5 (C) 80 μg/mL HEG-5 (D)

Fig. 3. Scanning electron micrographs of SGC-7901 showing morphological changes of cells treated without (A) and with 20 μg/mL (B), 50 μg /mL (C), and 80 μg/mL (D) of HEG-5 for 48 h. The surface of untreated cells (control) had numerous intact folds, while HEG-5-treated SGC-7901 cells showed the typical apoptotic characteristics including loss of folds, appearance of blebbing (20 and 50 μg/mL), and formation of apoptotic body (80μg/mL).

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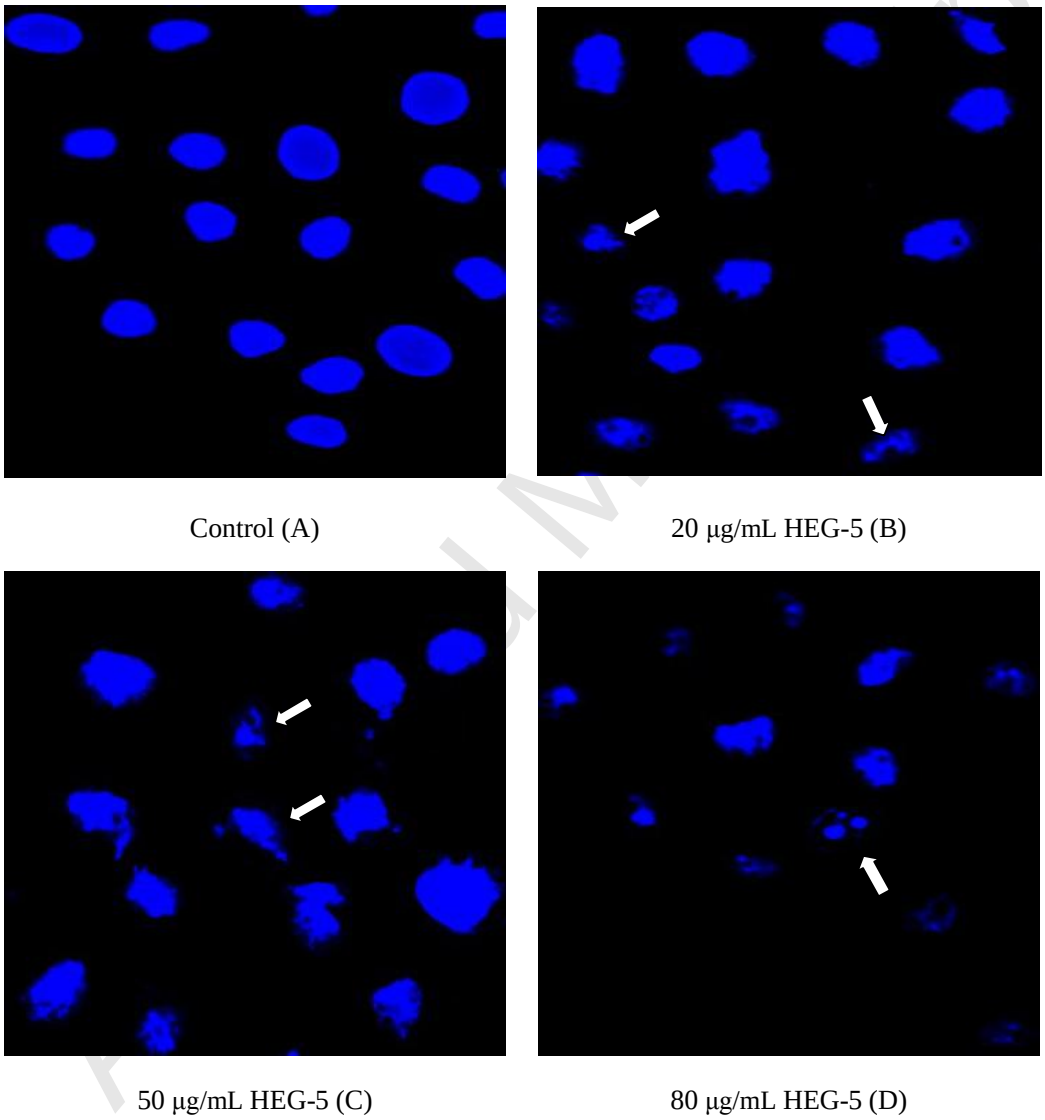

Fig. 4. Fluorescence micrographs of SGC-7901 cells stained with Hoechst 33258 (×100). Compared to the round and homogeneously stained nuclei in the control group, the nucleus of SGC-7901 cells treated by polysaccharide-protein HEG-5 were fragmented.

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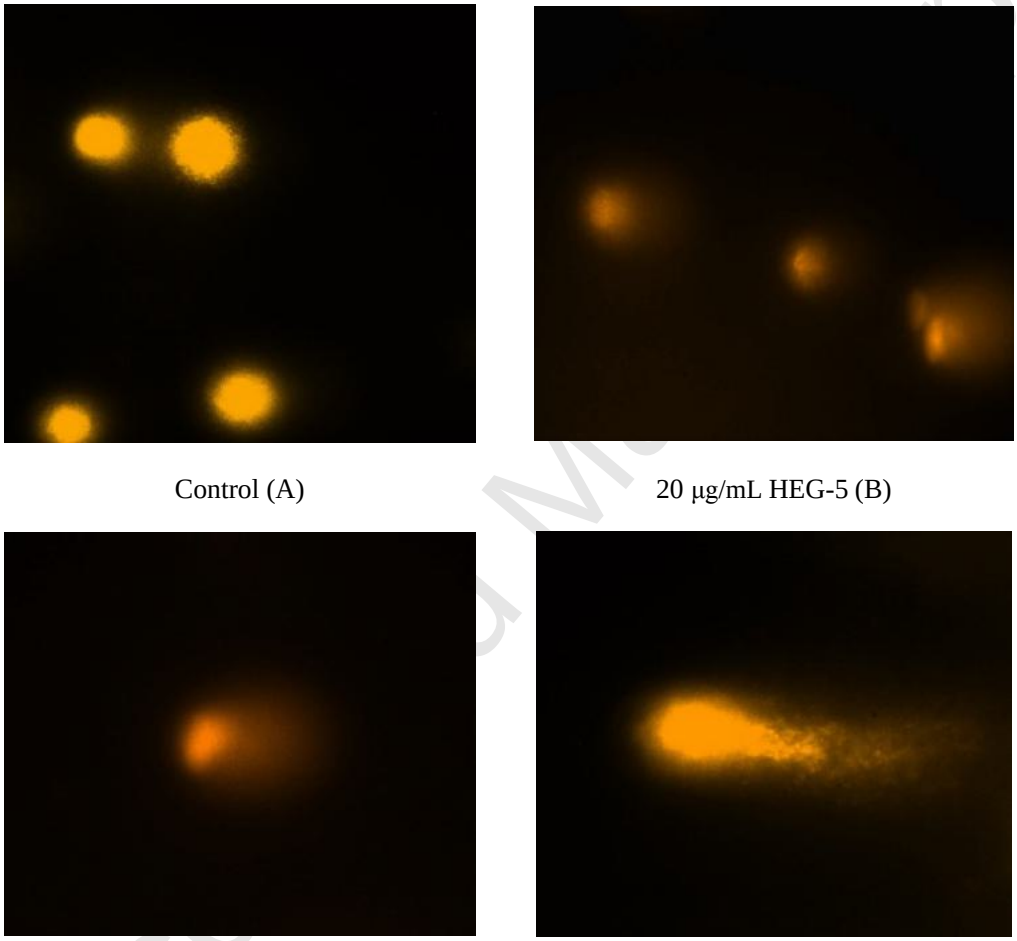

Fig.5. Comet assay micrographs of SGC-7901 treated with HEG-5 stained with ethidium bromide solution. The untreated SGC-7901 cells had the concentrated shape at the comet heads with high-density DNA accompanied with the smooth margin and intact nuclei. After treatment of HEG-5 at 50 μg/mL (C) and 80 μg/mL (D), the clear comet-like tails of DNA migration could be observed.

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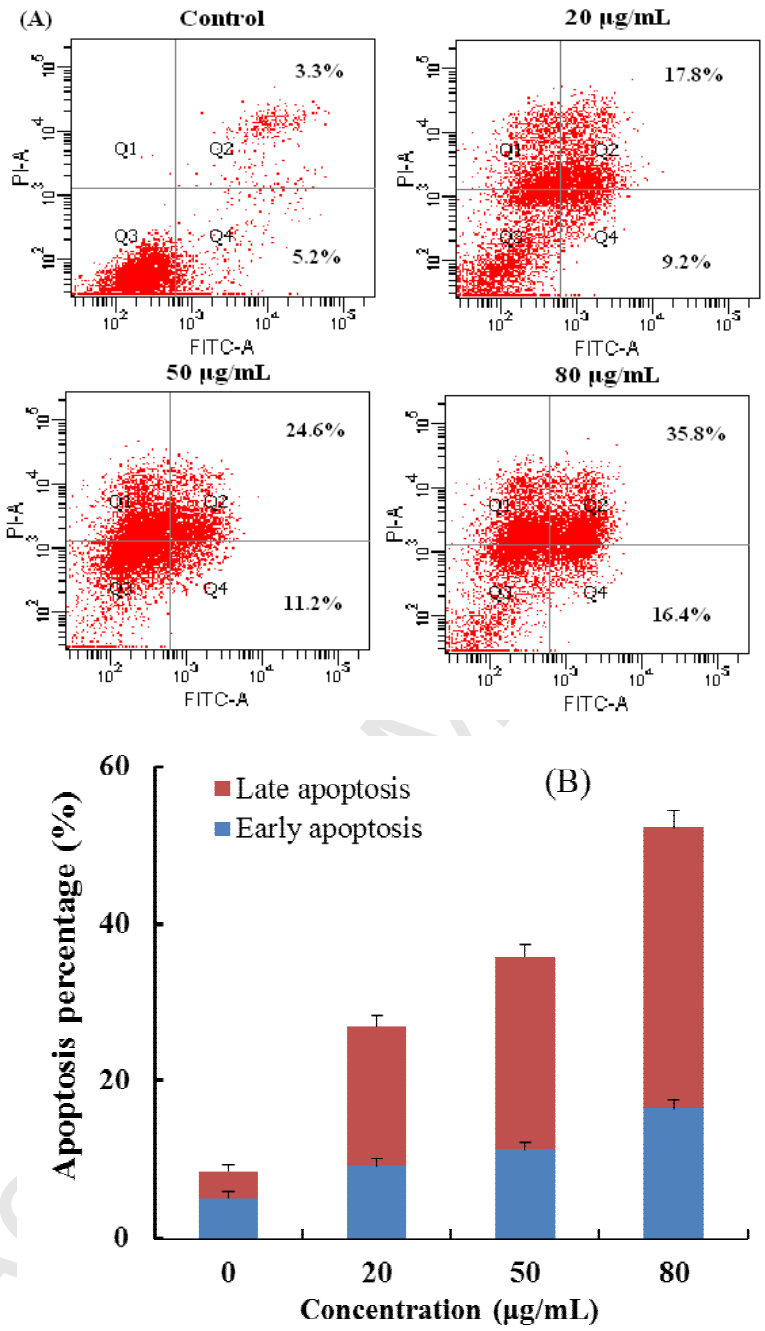

Fig.6. Quantitative analysis of apoptosis cells induced by HEG-5 measured by Annexin V-FITC and PI double staining method. (A) Representative dot plots of Annexin V/PI staining. (B) Column bar graph of apoptotic cells. Cells were treated with 50 and 80μg/mL of HEG-5 for 48 h and then analyzed by flow cytometry.

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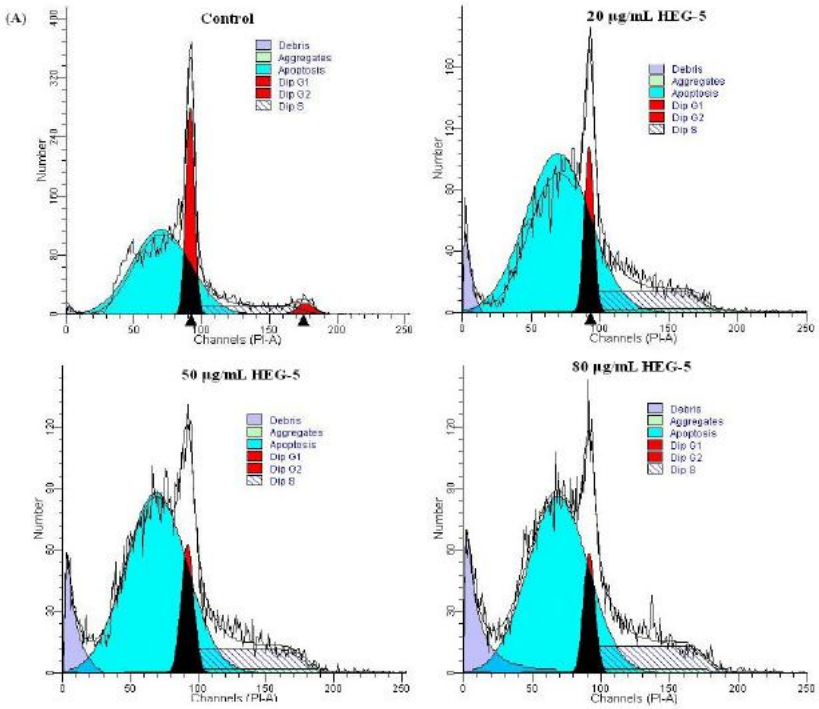

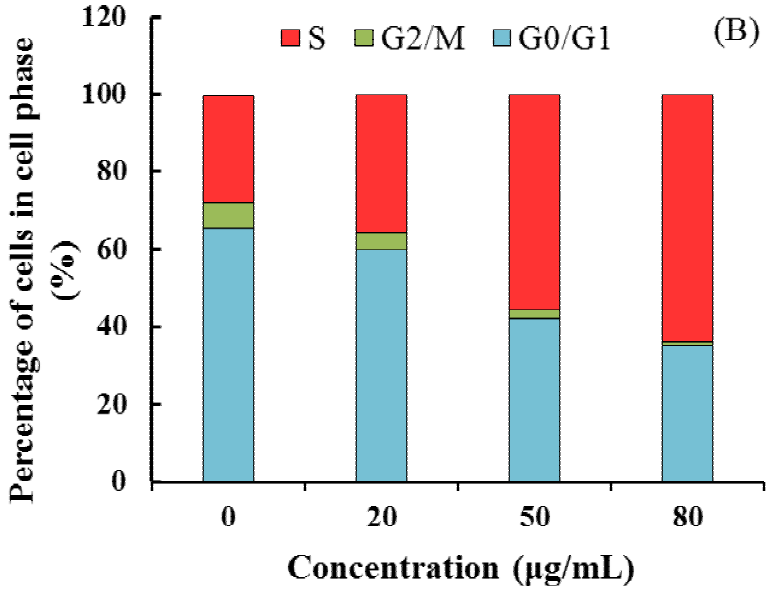

Fig. 7. Effects of HEG-5 on cell cycle phase distribution of SGC-7901 cells. (A) Representative histograms of DNA content in the cells incubated with HEG-5 at 0, 20, 50 and 80 μg/mL for 48 h. (B) Percentage of cell population in S, G2/M and G0/G1 phase.

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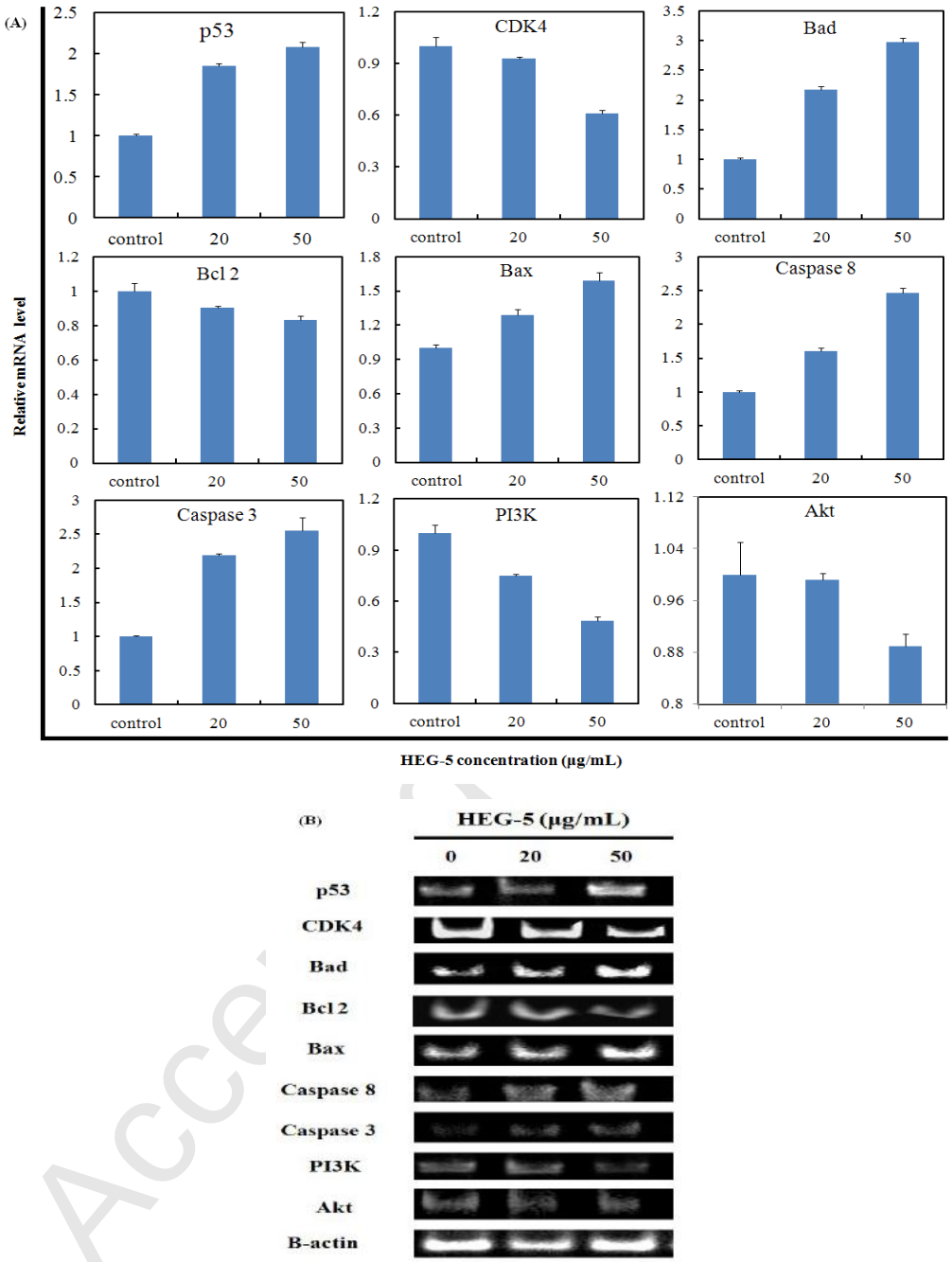

Fig. 8. Quantitaive real-time PCR analysis of p53, CDK4, Caspase 8, Caspase 3, Bcl 2, Bax, Bad, PI3k and AkT1 gene expression in SGC-7901 cells exposed to HEG-5 (0, 20, and 50 μg/mL) for 24 h.

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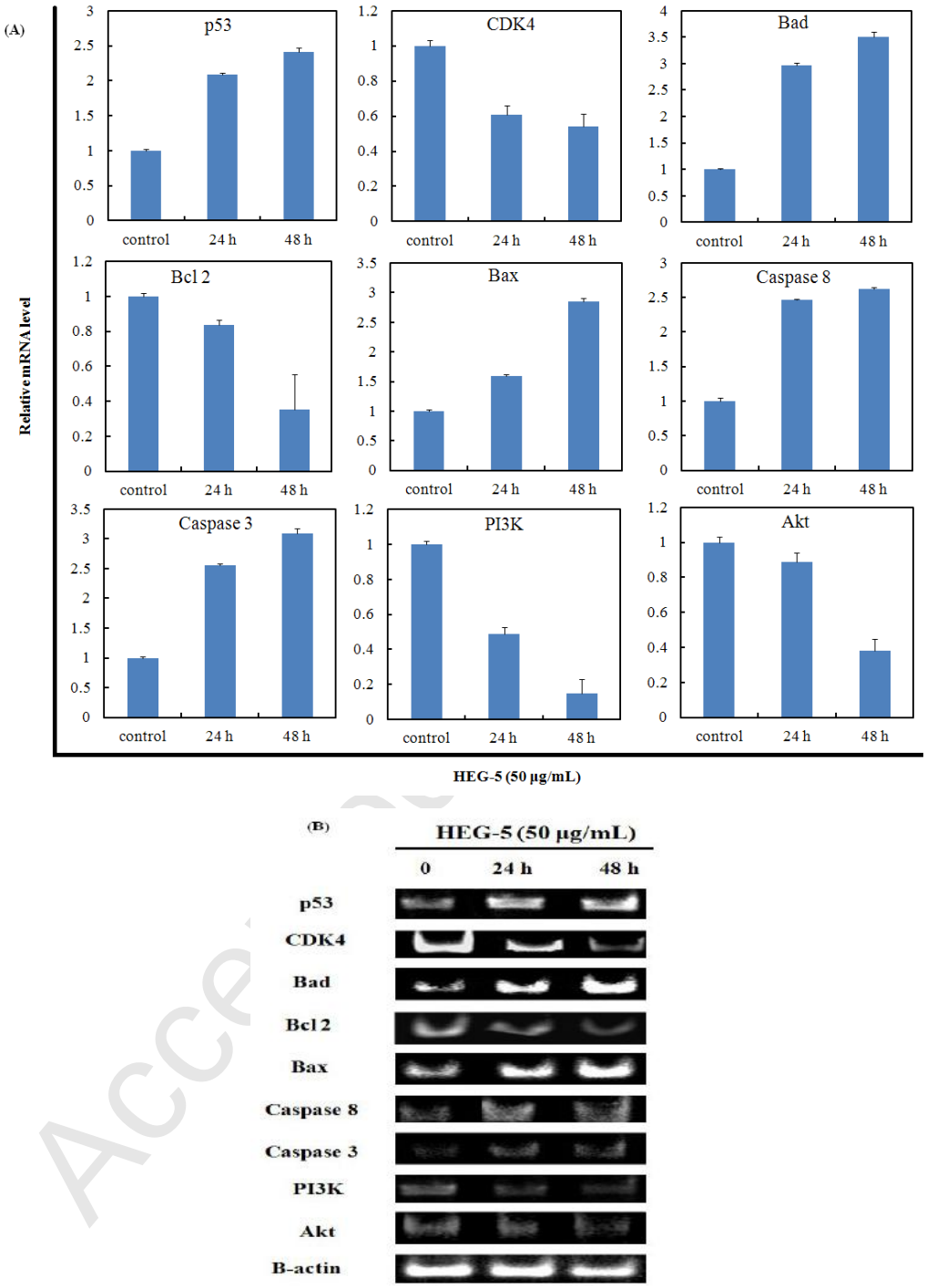

Fig. 9. Quantitative real-time PCR analysis of p53, CDK4, Caspase 8, Caspase 3, Bcl 2, Bax, Bad, PI3k and AkT1 gene expression in SGC-7901 cells exposed to HEG-5 (50 μg/mL) for 24 h and 48 h.

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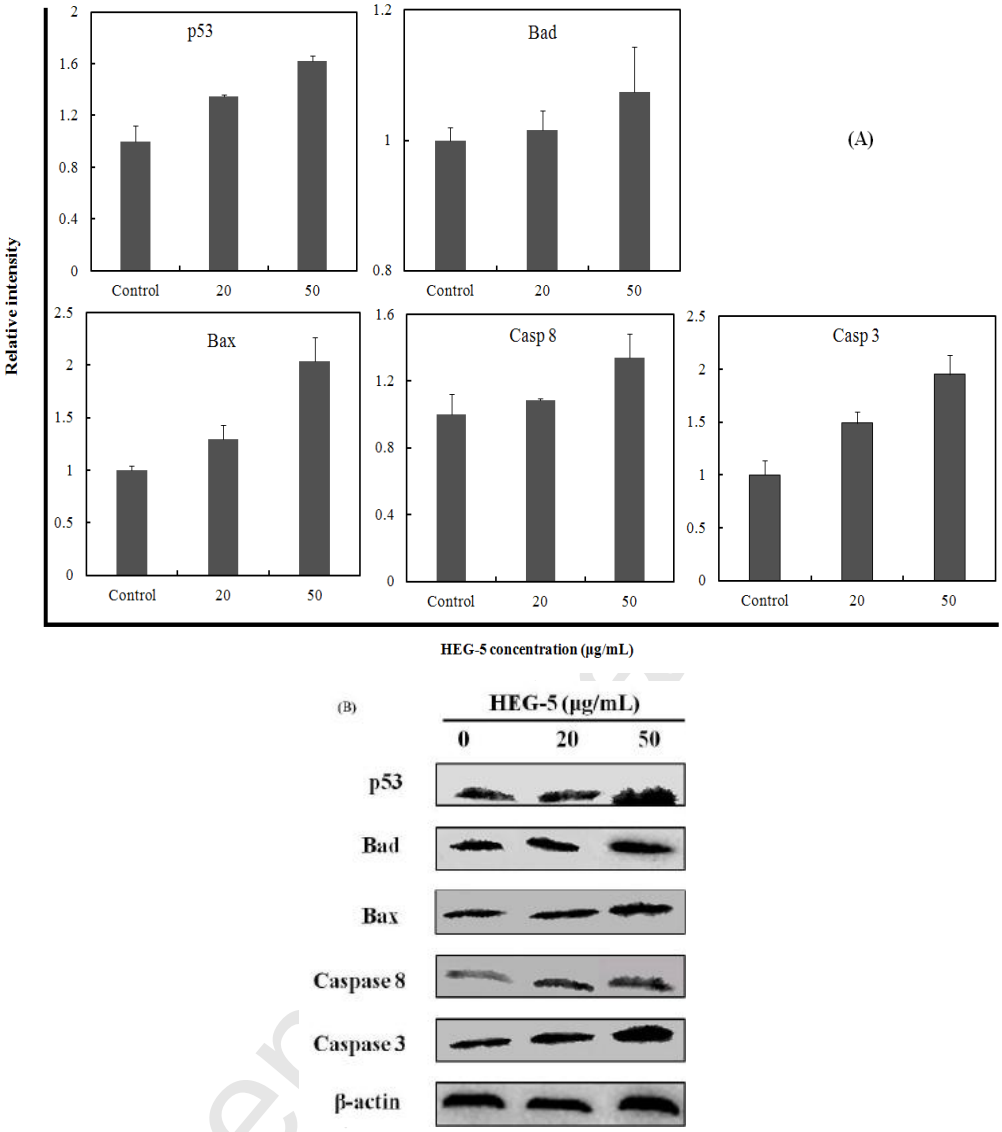

Fig. 10. Western blot for p53, Bad, Bax, Caspase 8, and Caspase 3 in SGC-7901 cells exposed to HEG-5 at 0, 20 and 50 μg/mL for 48 h.

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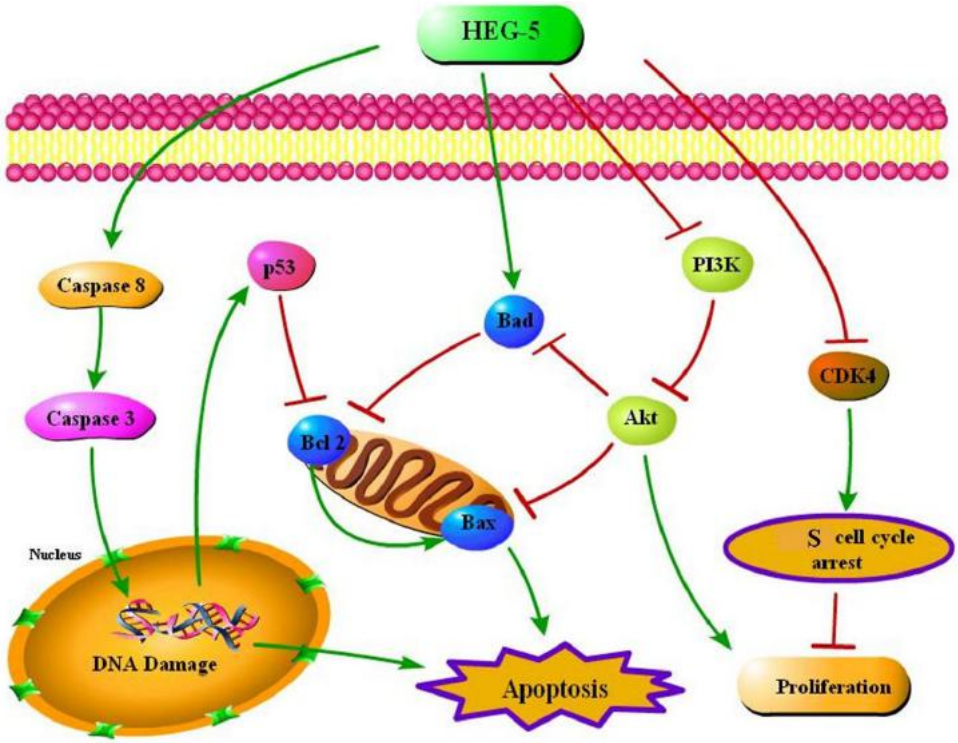

Fig. 11. Possible schematic diagram of antitumor mechanism of HEG-5 via the caspase-8/-3 dependent pathway, p53-dependent mitochondrial mediation, and PI3k/Akt regulation in SGC-7901 cells.